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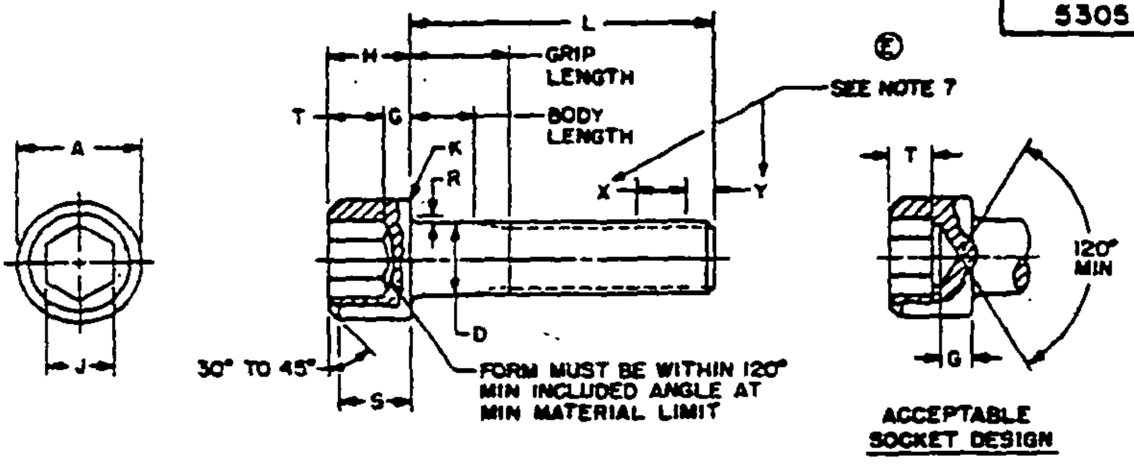

DIVISIONS: AT, AV, EL, IL, MI, NI, NSA, SH
USER: [illegible]

TABLE I

NOMINAL SIZE		(#0) .060	(#6) .138	(#8) .164	(#10) .190	(1/4) .250	(5/16) .3125
THREADS PER INCH UNF-3A		80	40	36	32	28	24
D BODY DIAMETER	MAX MIN	.0600 .0568	.1380 .1329	.1640 .1585	.1900 .1840	.2500 .2435	.3125 .3053
A HEAD DIAMETER	MAX MIN	.096 .091	.226 .218	.270 .262	.3125 .303	.3750 .365	.4688 .457
H HEAD HEIGHT	MAX MIN	.060 .057	.138 .134	.164 .159	.190 .185	.250 .244	.3125 .306
S HEAD SIDE HGT	MIN	.054	.124	.148	.171	.225	.281
J SOCKET WIDTH ACROSS FLATS	MAX MIN	.0510 .0500	.1111 .1094	.1426 .1406	.1587 .1562	.1900 .1875	.2530 .2500
T- KEY ENGAGEMENT	MIN	.025	.064	.077	.090	.120	.151
G WALL THICKNESS	MIN	.020	.047	.056	.065	.095	.119
K CHAMFER OR RADIUS	MAX	.003	.005	.005	.005	.008	.008
R FILLET EXTENSION	MAX MIN	.007 .001	.010 .006	.012 .007	.014 .009	.014 .009	.017 .012
• TENSILE STRENGTH LBS - MIN		270	1,570	2,290	3,110	5,600	9,100

L LENGTH	TOLERANCE	(#0) DASH NO.	(#6) DASH NO.	(#8) DASH NO.	(#10) DASH NO.	(#10) GRIP LENGTH MAX	(#10) BODY LENGTH MIN	(1/4) DASH NO.	(1/4) GRIP LENGTH MAX	(1/4) BODY LENGTH MIN	(5/16) DASH NO.	(5/16) GRIP LENGTH MAX	(5/16) BODY LENGTH MIN
.125	+.000 -.031	1											
.188		2											
.250		3	9										
.375		4	10	18	26			40					
.500		5	11	19	27			41			55		
.625			12		28			42			56		
.750			13	21	29			43			57		
.875					30			44			58		
1.000					31			45			59		
1.250	+.000 -.040				32	.375	.167	46			60		
1.500					33	.375	.167	47	.500	.250	61		
1.750					34	.875	.667	48	.500	.250	62	.625	.347
2.000					35	.875	.667	49	1.000	.750	63	.625	.347
2.250								50	1.000	.750	64	1.125	.847
2.500											65	1.125	.847

For notes see sheet 2.

Ⓔ For changes see sheets 1, thru 5.

APPROVED 5 APR 1962 REVISED Ⓐ 21 FEB 1963 Ⓑ 18 JUL 1963 Ⓒ 25 MAR 1965 Ⓓ 22 OCT 1970 Ⓔ 31 DEC 1976

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document, when applicable.

P.A. WC Other Cust AS 99	TITLE Ⓔ SCREW,CAP,SOCKET HEAD AND SCREW,CAP,SOCKET HEAD,SELF-LOCKING: ALLOY STEEL,CADMIUM PLATED,UNF-3A	MILITARY STANDARD MS16998
PROCUREMENT SPECIFICATION FF-S-86	SUPERSEDES BCMX3, 3.1, 3.2; BCMX4 and 4.2 MS35456, MS35458 and MS35460 in part	SHEET 1 OF 5

DD FORM 1 OCT 64 672-1 (Continued) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. 5305-1185

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REVIEWER: AT, AV, EA, IS, MI, MU, NSA, OS
USER: ME, PA, WT, YD

TABLE I (Cont'd)

NOMINAL SIZE		(3/8) .375	(7/16) .4375	(1/2) .500	(5/8) .625
THREADS PER INCH UNF-3A		24	20	20	18
D BODY DIAMETER	MAX	.3750	.4375	.5000	.6250
	MIN	.3678	.4294	.4919	.6163
A HEAD DIAMETER	MAX	.5625	.6562	.750	.9375
	MIN	.550	.642	.735	.921
H HEAD HEIGHT	MAX	.375	.4375	.500	.625
	MIN	.368	.430	.492	.616
S HEAD SIDE HGT	MIN	.337	.394	.450	.562
J SOCKET WIDTH ACROSS FLATS	MAX	.3160	.3790	.3790	.5050
	MIN	.3125	.3750	.3750	.5000
T KEY ENGAGEMENT	MIN	.182	.213	.245	.307
C WALL THICKNESS	MIN	.143	.166	.190	.238
E CHAMFER OR RADIUS	MAX	.008	.010	.010	.010
F FILLET EXTENSION	MAX	.020	.023	.026	.032
	MIN	.015	.018	.020	.024
* TENSILE STRENGTH LBS - MIN		13,800	18,700	25,200	40,400

LENGTH	TOLERANCE SIZE .375	TOLERANCE OVER .375	DASH NO.	GRIP LENGTH MAX	BODY LENGTH MIN	Ⓔ→ DASH NO.	GRIP LENGTH MAX	BODY LENGTH MIN	DASH NO.	GRIP LENGTH MAX	BODY LENGTH MIN	DASH NO.	GRIP LENGTH MAX	BODY LENGTH MIN
.500	+.000 −.031	+.000 −.031	70											
.625			71											
.750			72											
.875			73											
1.000			74			~~86~~			95			106		
1.250	+.000 −.040	+.000 −.060	75			~~87~~			96			107		
1.500			76			~~88~~			97			108		
1.750			77	.500	.187				98			109		
2.000			78	.500	.187	~~89~~	.625	.268	99			110		
2.250			79	1.000	.687				100	.750	.365	111		
2.500			80	1.000	.687				101	.750	.365	[illegible]	.750	.295
3.000	−.060	−.080	81	1.500	1.187				102	1.500	1.115	[illegible]	.750	.295

NOTES:

1. MATERIAL: Alloy steel in accordance with procurement specification.
2. PROTECTIVE COATING: Cadmium plated in accordance with procurement specification.
3. THREADS: The threads shall be in accordance with procurement specification.
4. THREAD LENGTH: Screws above heavy line shall have complete (full form) threads extending within two threads of the head as measured with a thread ring gauge.
5. DIMENSIONS: All dimensions are in inches unless otherwise specified.
6. HEAD: The head may or may not be knurled, at the manufacturer's option. Type of knurling optional.

Ⓔ7. SELF-LOCKING ELEMENT:
   (a) The self-locking element shall be a patch type or longitudinal strip in accordance with MIL-F-18240.
   (b) For "X" and "Y" dimensions, design and usage limitations, see MS15981.
   (c) Maximum ring gage diameter that the locking region of screw must pass thru freely or with finger pressure shall be the nominal size plus 0.010.

8. *TENSILE STRENGTH: Based on 160,000 PSI minimum ultimate tensile strength. Load pounds are calculated by the stress areas indicated in the Screw-Thread Standards for Federal Services, Handbook H28.

Ⓔ9. PART NUMBER: The MS part number consists of the MS number, plus the dash number. For screws with the self-locking element add the letter "L" after the dash number.
Example: MS16998-1 Plain Screw
MS16998-1L Self-locking Screw

10. Referenced documents shall be of the issue in effect on date of invitations for bid.
11. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

Ⓔ ** INACTIVE FOR NEW DESIGN AFTER 31 DECEMBER 1976.

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document, when applicable.

APPROVED 5 APR 1962   REVISED Ⓔ for changes see sheets 1 thru 5

P.A. WC Other Cust AS 99	TITLE Ⓔ SCREW,CAP,SOCKET HEAD AND SCREW,CAP,SOCKET HEAD,SELF-LOCKING: ALLOY STEEL,CADMIUM PLATED,UNF-3A	MILITARY STANDARD MS16998
PROCUREMENT SPECIFICATION FF-S-86	SUPERSEDES: BCM23, 3.1, 3.2; BCM24 and 4.2 MS35456, MS35458 and MS35460 in part	SHEET 2 OF 5

DD FORM 672-1 (Limited reproduction)   [illegible]

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REVIEWER: AT, AV, LA, IS, NI, MU, MSA, OS
USER: ME, PA, MW, YD

TABLE II
INTERCHANGEABILITY TABLE

The screws covered by dash numbers given in MS35456, MS35458 and MS35460 that are replaced by applicable dash numbers in this document are cancelled after 5 April 62. Use the dash numbers given in the preceding sheets. The cancelled screws cannot always replace the new screws and should be used until existing stocks are depleted. Use only the new screws for design and replacement. Replacement shall be in accordance with this table. Interchangeability information regarding part numbers on superseded drawings are also contained in this table.

PART NUMBERS		PART NUMBERS		PART NUMBERS	
CANCELLED	SUPERSEDING	CANCELLED	SUPERSEDING	CANCELLED	SUPERSEDING
MS35456 MS35458 MS35460	MS16998	MS35456 MS35458 MS35460	MS16998	MS35456 MS35458 MS35460	MS16998
Dash No.	Dash No.	Dash No.	Dash No.	Dash No.	Dash No.
1	9	22	42	53	61
2	10	23	43	62	95
3	11	24	44	63	96
4	12	25	45	64	97
5	13	26	46	65	98
6	18	27	47	66	99
7	19	28	48	67	100
9	21	29	49	68	101
10	26	30	50	69	102
11	27	42	70		
12	28	43	71		
13	29	44	72		
14	30	45	73		
15	31	46	74		
16	32	47	75		
17	33	48	76		
18	34	49	77		
19	35	50	78		
20	40	51	79		
21	41	52	80		

Screws covered by the following part numbers are inactive for Design and Engineering. They shall be used for maintenance of existing equipment to replace the 5/16, 7/16 and 5/8 inch size screws used in counterbored holes or other close applications.

PART NO.	PART NO.	PART NO.	PART NO.	PART NO.	PART NO.
INACTIVE	INACTIVE	INACTIVE	INACTIVE	INACTIVE	INACTIVE
MS35456 MS35458 MS35460	MS35456 MS35458 MS35460	MS35456 MS35458 MS35460	MS35456 MS35458 MS35460	MS35456 MS35458 MS35460	MS35456 MS35458 MS35460
Dash No.	Dash No.	Dash No.	Dash No.	Dash No.	Dash No.
31	35	39	55	71	75
32	36	40	56	72	76
33	37	41	58	73	77
34	38	54	70	74	

APPROVED 5 APR 1962 REVISED (E) for changes see sheets 1 thru 5

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document, when applicable.

P.A. WC Other Cust AS 99	TITLE (E) SCREW, CAP, SOCKET HEAD AND SCREW, CAP, SOCKET HEAD, SELF-LOCKING: ALLOY STEEL, CADMIUM PLATED, UNF-3A	MILITARY STANDARD	MS16998
PROCUREMENT SPECIFICATION FF-S-86	SUPERSEDES: BCVX1 3.1, 3.2; BCVX4 and 4.2 MS35456, MS35458 and MS35460 in part	SHEET 3 OF 5	

DD FORM 672-1 (Continued) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

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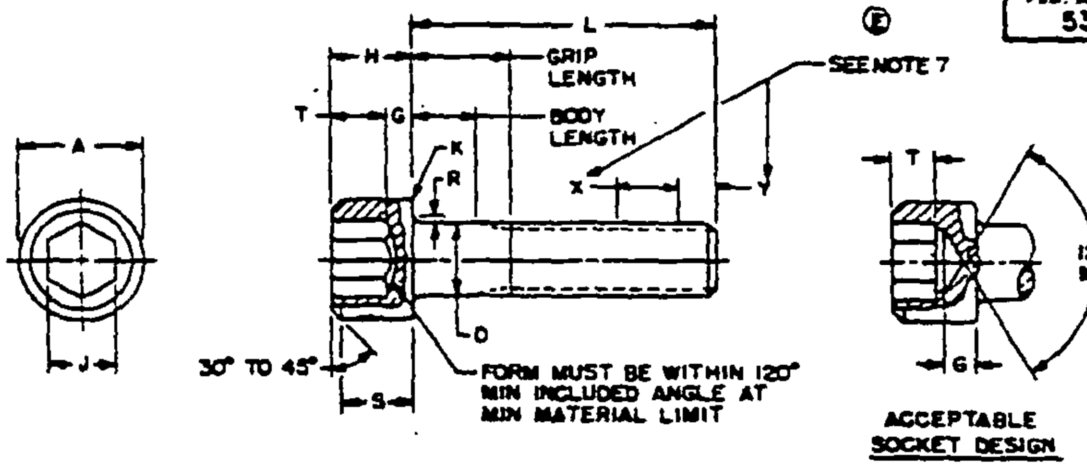

TABLE III
METRIC TRANSLATION OF TABLE I

NOMINAL SIZE mm		1.524	3.505	4.166	4.826	6.350	7.938
THREADS PER 25.4 mm UNF-3A		80	40	36	32	28	24
D BODY DIAMETER	MAX MIN	1.524 1.443	3.505 3.376	4.166 4.026	4.826 4.674	6.350 6.185	7.938 7.755
A HEAD DIAMETER	MAX MIN	2.438 2.311	5.740 5.537	6.858 6.655	7.938 7.696	9.550 9.271	11.908 11.608
H HEAD HEIGHT	MAX MIN	1.524 1.448	3.505 3.404	4.166 4.039	4.826 4.699	6.350 6.198	7.938 7.772
S HEAD SIDE HGT	MIN	1.372	3.150	3.759	4.343	5.715	7.137
J SOCKET WIDTH ACROSS FLATS	MAX MIN	1.295 1.270	2.822 2.779	3.622 3.571	4.031 3.967	4.826 4.762	6.426 6.350
T KEY ENGAGEMENT	MIN	.635	1.626	1.956	2.286	3.048	3.835
G WALL THICKNESS	MIN	.508	1.194	1.422	1.651	2.413	3.023
K CHAMFER OR RADIUS	MAX	.076	.127	.127	.127	.203	.203
R FILLET EXTENSION	MAX MIN	.178 .076	.254 .152	.305 .178	.356 .229	.356 .229	.432 .305
* TENSILE STRENGTH Ⓔ kN - MIN		1.215	7.065	10.305	13.995	25.360	40.950

L LENGTH	TOLERANCE	DASH NO.	DASH NO.	DASH NO.	DASH NO.	GRIP LENGTH MAX	BODY LENGTH MIN	DASH NO.	GRIP LENGTH MAX	BODY LENGTH MIN	DASH NO.	GRIP LENGTH MAX	BODY LENGTH MIN
3.175	+.000 -.787	1											
4.775		2											
6.350		3	9										
9.525		4	10	18	26			40					
12.700		5	11	19	27			41			55		
15.875			12		28			42			56		
19.050			13	21	29			43			57		
22.225					30			44			58		
25.400					31			45			59		
31.750	+.000 -1.016				32	9.525	4.242	46			60		
38.100					33	9.525	4.242	47	12.700	6.350	61		
44.450					34	22.225	16.942	48	12.700	6.350	62	15.875	8.814
50.800					35	22.225	16.942	49	25.400	19.050	63	15.875	8.814
57.150								50	25.400	19.050	64	28.575	21.514
63.500											65	28.575	21.514

NOTE: All dimensions are in millimeters (mm), unless otherwise specified and are based on an approximate conversion factor of 25.4mm equal one inch.

REVIEWERS: AT, AV, EA, IS, MI, MO, MSA, OS
USERS: ME, PA, WP, YD

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APPROVED 5 APR 1982 REVISED Ⓔ for changes see sheets 1 thru 5

P.A. WC Other Cust AS 99	TITLE Ⓔ SCREW, CAP, SOCKET HEAD AND SCREW, CAP, SOCKET HEAD, SELF-LOCKING: ALLOY STEEL, CADMIUM PLATED, UNF-3A	MILITARY STANDARD	MS16998
PROCUREMENT SPECIFICATION FF-S-86	SUPERSEDES: BCXX3, 3.1, 3.2; BCXX4 and 4.2 MS35456, MS35458 and MS35460 in part	SHEET 4 OF 5	

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TABLE III (Cont'd)
METRIC TRANSLATION OF TABLE I

NOMINAL SIZE mm	9.525	11.112	12.700	15.875
THREADS per 25.4mm UNF-3A	24	20	20	18
D BODY DIAMETER MAX MIN	9.525 9.342	11.112 10.907	12.700 12.494	15.875 15.654
A HEAD DIAMETER MAX MIN	14.288 13.970	16.667 16.307	19.050 18.669	23.812 23.393
H HEAD HEIGHT MAX MIN	9.525 9.347	11.112 10.922	12.700 12.497	15.875 15.646
S HEAD SIDE HGT MIN	8.560	10.008	11.430	14.275
J SOCKET WIDTH ACROSS FLATS MAX MIN	8.026 7.938	9.627 9.525	9.627 9.525	12.827 12.700
T KEY ENGAGEMENT MIN	4.623	5.410	6.223	7.798
G WALL THICKNESS MIN	3.632	4.216	4.826	6.045
K CHAMFER OR RADIUS MAX	.203	.254	.254	.254
R FILLET EXTENSION MAX MIN	.508 .381	.584 .457	.660 .508	.813 .610
• TENSILE STRENGTH kN - MIN Ⓔ	62.100	84.150	113.400	181.800

L LENGTH	TOLERANCE		DASH NO.	GRIP LENGTH MAX	BODY LENGTH MIN	Ⓔ ⇒ DASH NO.	GRIP LENGTH MAX	BODY LENGTH MIN	DASH NO.	GRIP LENGTH MAX	BODY LENGTH MIN	DASH NO.	GRIP LENGTH MAX	BODY LENGTH MIN
	SIZE 9.525	OVER 9.525												
12.700	+.000 -.787	+.000 -.787	70											
15.875			71											
19.050			72											
22.225			73											
25.400			74			~~86~~			95			106		
31.750	+.000 -1.016	+.000 -1.524	75			~~87~~			96			107		
38.100			76			~~88~~			97			108		
44.450			77	12.700	4.750				98			109		
50.800			78	12.700	4.750	~~89~~	15.875	6.807	99			110		
57.150			79	25.400	17.450				100	19.050	9.271	111		
63.500			80	25.400	17.450				101	19.050	9.271	112	19.050	7.493
76.200	-1.524	-2.032	81	38.100	30.150				102	38.100	28.321	113	19.050	7.493

NOTE: All dimensions are in millimeters (mm), unless otherwise specified and are based on an approximate conversion factor of 25.4mm equal one inch.

Ⓔ ** INACTIVE FOR NEW DESIGN AFTER 31 DECEMBER 1976.

REVIEWER: AT, AV, EA, IS, MI, MU, MSA, OS
USER: ME, PA, WL, YD

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APPROVED 5 APR 1962 REVISED Ⓔ For changes see sheets 1 thru 5

☆ U. S. GOVERNMENT PRINTING OFFICE: 1977-703-020/1079

P.A. WC Other Cust AS 99	TITLE Ⓔ SCREW, CAP, SOCKET HEAD AND SCREW, CAP, SOCKET HEAD, SELF-LOCKING: ALLOY STEEL, CADMIUM PLATED, UNF-3A	MILITARY STANDARD MS16998
PROCUREMENT SPECIFICATION FF-S-86	SUPERSEDES BCWI3, 3.1, 3.2; BCWI4 and 4.2 MS35456, MS35458 and MS35460 in part.	SHEET 5 OF 5

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.